

Clinical Trial Subject Recruitment Engine

1. Problem Context & Overview

Medical research institutions require an eligibility pre-screening platform for clinical trials, tracking subject inclusion/exclusion criteria, informed consent milestones, and visit compliance logs.

Target Stakeholders / Actors: Trial Candidate, Clinical Researcher

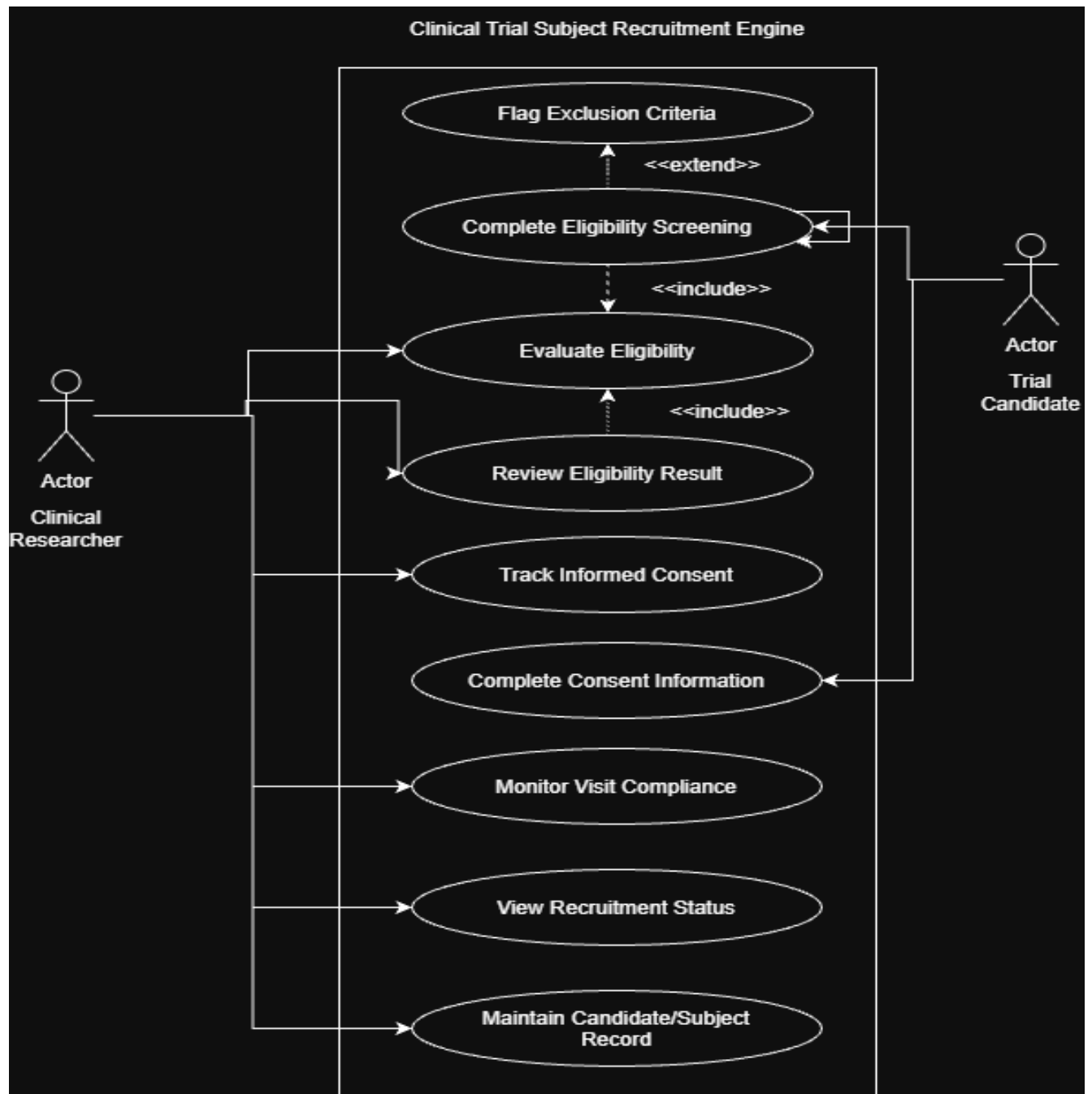
2. Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	Candidate eligibility evaluation against inclusion/exclusion criteria	High	Pass: Ineligible candidates are flagged with the failed criteria. Fail: A candidate with an exclusion criterion is marked eligible.	Ensures only protocol-eligible candidates proceed.
FR-002	Functional	The system shall record and track informed-consent milestones for each candidate.	High	Pass: Consent status and milestone date/time are recorded and viewable. Fail: A candidate cannot be incorrectly shown as fully consented when a required milestone is incomplete.	Ensures informed consent is properly documented before trial participation.

FR-003	Functional	The system shall allow clinical researchers to review candidate eligibility results and recruitment status.	High	Pass: Researcher can view eligibility result, failed criteria, consent status, and recruitment status. Fail: Researcher cannot access or sees incomplete recruitment information.	Supports researchers in making informed recruitment decisions.
FR-004	Functional	The system shall maintain subject visit compliance logs against the trial's scheduled visits.	Medium	Pass: Completed, missed, and upcoming visits are accurately recorded. Fail: A missed visit is incorrectly recorded as completed.	Helps researchers monitor participant adherence to the trial protocol.
FR-005	Functional	The system shall generate recruitment and subject-status records for clinical researchers.	Medium	Pass: Researcher can obtain current candidate/subject status and recruitment information. Fail: Generated records contain missing or inconsistent subject statuses.	Provides researchers with reliable recruitment monitoring information.
NFR-001	Non-Functional — Performance & Security	The recruitment database shall comply with applicable 21 CFR Part 11 electronic-record requirements and maintain secure, tamper-evident audit trails.	High	Pass: Security/compliance testing demonstrates authenticated access, traceable record changes, and tamper-evident audit records under expected operating conditions. Fail: Unauthorized or untraceable modification of regulated records is possible.	Clinical-trial records require integrity, traceability, and regulatory compliance.

NFR -002	Non-Functional — Usability & Reliability	The system shall provide a reliable and easy-to-use interface for eligibility screening, consent tracking, and visit compliance monitoring.	High	Pass: Representative users can complete core screening and tracking tasks without system errors, and the system maintains data correctly during normal operation. Fail: Core tasks routinely result in errors, lost data, or confusing workflows.	Reduces researcher workload and minimizes errors in subject recruitment and tracking.
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2. UML Use-Case Diagram



3. Use-Case Flow Specification

Title

Use-Case Flow Specification — UC-001: Complete Eligibility Screening

1. Use-Case ID and Name

- **ID:** UC-001
- **Name:** Complete Eligibility Screening
- **Primary Actor:** Trial Candidate
- **Supporting Actor:** Clinical Researcher

2. Goal

State that the purpose is to determine whether a trial candidate satisfies the protocol's inclusion and exclusion criteria before recruitment.

3. Preconditions

Include things such as:

- The clinical trial is configured in the system.
- Inclusion and exclusion criteria have been entered.
- The candidate has access to the screening process.
- The candidate has provided the required identifying information.

4. Postconditions

Include:

- Candidate responses are stored securely.
- Eligibility has been evaluated.
- The candidate's eligibility status is recorded.
- Any failed/exclusion criteria are identified.
- The clinical researcher can review the screening result.

5. Main Success Scenario

Use numbered steps:

1. Trial Candidate starts the eligibility screening.
2. System displays the required screening questions.
3. Candidate enters responses.

4. System validates the submitted responses.
5. System compares the responses against the trial's inclusion and exclusion criteria.
6. System determines the candidate's eligibility status.
7. System records the screening result and applicable criteria.
8. System makes the result available to the Clinical Researcher.
9. System displays the candidate's eligibility status.

6. Alternate Flow

Use one clear alternate flow, for example:

A1. Candidate fails an exclusion criterion

1. During evaluation, the system detects a response matching an exclusion criterion.
2. The system marks the candidate as **Ineligible**.
3. The system records the specific failed/exclusion criterion.
4. The system prevents the candidate from being incorrectly classified as eligible.
5. The Clinical Researcher can review the reason for ineligibility.

7. Related Requirement

Related Requirement: FR-001 — Eligibility evaluation against trial inclusion/exclusion criteria.